



JOSEPH AYO BABALOLA UNIVERSITY
(Established By Christ Apostolic Church, World Wide)
The First Entrepreneurial University in Nigeria
Km 36, Ilesa-Akure Road, Ikeji -Arakeji, Osun State.

DEPARTMENT OF PHYSICAL SCIENCES

PHY 107
GENERAL PRACTICAL PHYSICS I

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Learning Outcomes

At the end of the course, students should be able to:

1. conduct measurements of some physical quantities;
2. make observations of events, collect and tabulate data;
3. identify and evaluate some common experimental errors;
4. plot and analyse graphs; and
5. draw conclusions from numerical and graphical analysis of data.

Course Contents

This introductory course emphasizes quantitative measurements. Experimental techniques. The treatment of measurement errors. Graphical analysis. The experiments include studies of meters, mechanical systems, heat, etc. (covered in PHY 101, 103). However, emphasis should be placed on the basic physical techniques for observation, measurements, data collection, analysis, and deduction.

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CHAPTER ONE

REPORT WRITING

1.0 Forms of report

The written account of the experimental work will judge the quality of your experimental work. Therefore, it is very important that students should know the techniques of writing report. The write up must be done accurately and carefully. It must be few, concise and neat. This will make the report easier to read and grade.

The following format should always be used to report experimental physics

- **Date:** The date of experiment must be clearly stated
- **Experiment number and Title:** The number and title of experiment must appear as written in the manual
- **Aim or Objectives:** The aim or objectives must be stated clearly and must be concise
- **Apparatus:** The list of apparatus supplied and used in the laboratory must be spelled out
- **Procedure/Method:** The procedure used in carrying out the experiment must be reported in reporting speech. The use of 'I' or 'We' must be avoided. It should be written in past tense
- **Diagram:** A schematic diagram of the set-up used in the laboratory must be drawn and well labelled. This should be drawn with **pencil**.
- **Data:** The data involves the table of reading and the measurement taken. The measurement and the appropriate units should be recorded either in tabular form whenever possible or otherwise. Your data should be neat and not written roughly.
- **Graph:** The appropriate graph should be plotted if required
- **Calculation/ Question:** All calculations should be done. This includes slope, intercept and other relevant calculations and questions should be answered
- **Discussion:** Where appropriate, the experiment should be discussed
- **Precaution:** The precautions taken to get accurate results should be stated.

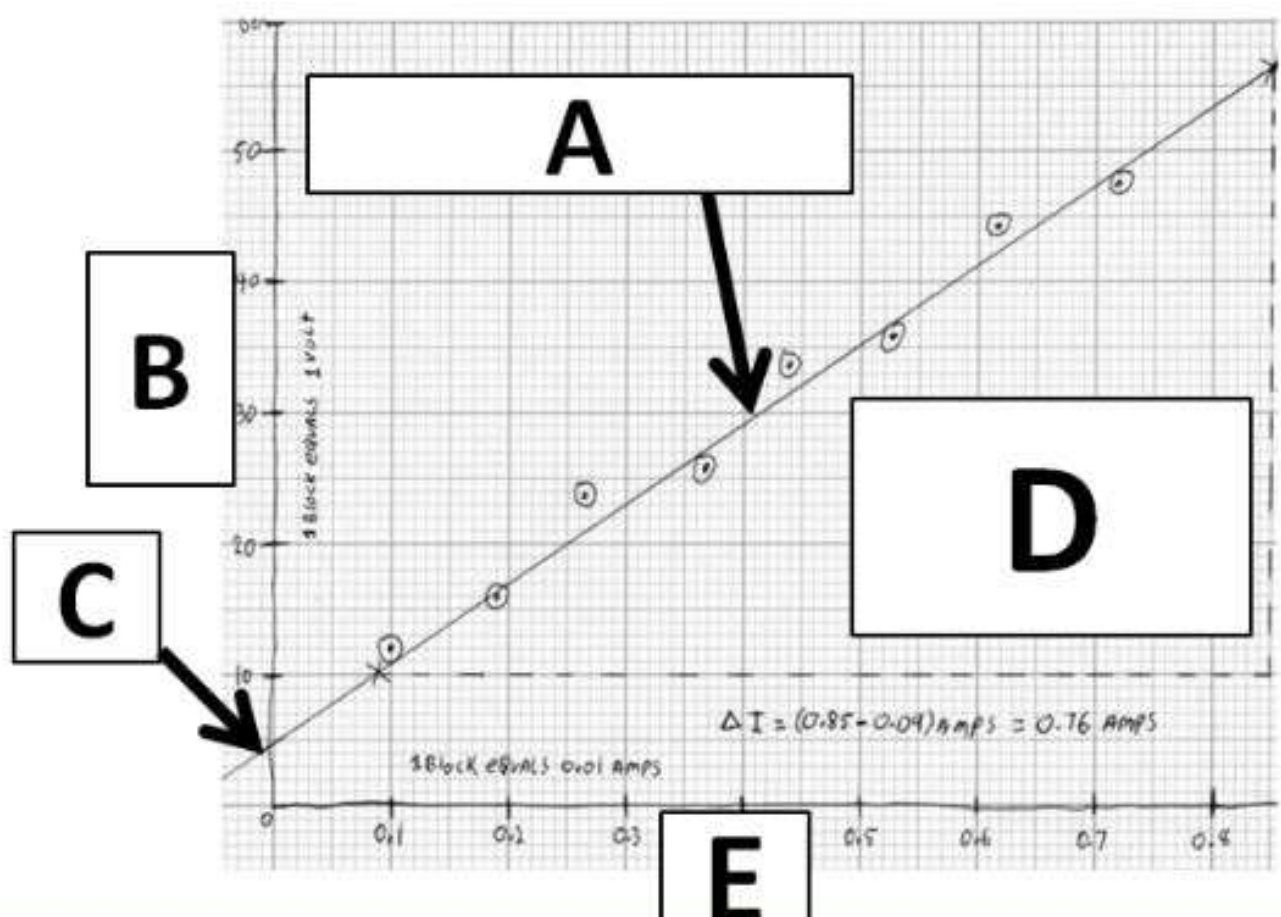
NOTE: *All works should be done inside the workbook (graph book)*

CHAPTER TWO

GRAPHING

2.0 Graph

A graph gives the relationship between two quantities. One is plotted against another on two axes at right-angles to each other. The quantity of the independent variable can be controlled while dependent quantity is determined by the outcome of the experiment or some mathematical relationship. We usually plot the independent variable on the x-axis which can also be called abscissa and the dependent variable on the y-axis which is the ordinate.



The following labels are used for the graph:

- A. Line of best Fit
- B. Ordinate (Y-axis)
- C. Intercept
- D. Slope
- E. Abscissa (X-axis)

2.1.1 Graphical Relationships

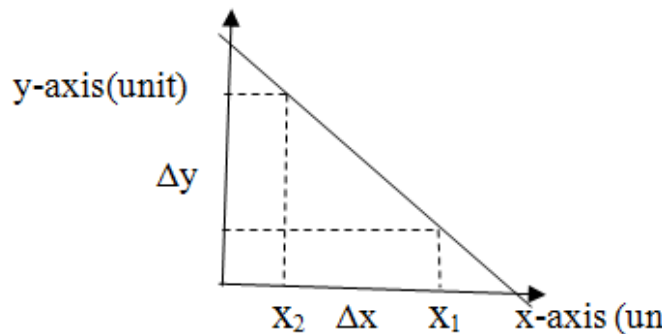
It can be straight line graph, curved graph which can be hyperbola, parabolic and exponential curve.

2.1.1 The Straight Line graph representation

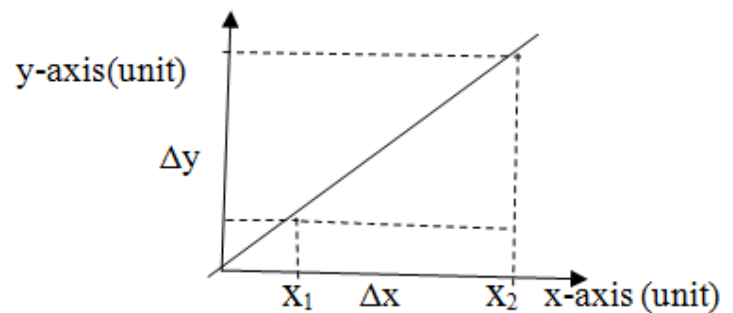
A straight line graph has an equation

$$y = mx + c \quad (1)$$

Where y is the dependent variable, m is the slope or gradient of the graph, x is the independent variable and c is the intercept. The gradient is determined by the change in y and change in x . example of a straight line graph is shown in figure below. It can start from origin and it may not.



This is a negative slope graph



This is a positive slope graph

Note: The point where the line touches X-axis and Y-axis is the **intercept** on x and y- axes respectively.

The slope(S) of the graph can be calculated using the formula:

$$m = \text{slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_1 - x_2}$$

$$m = \text{slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Where y and x are variables. It is not a fixed parameter

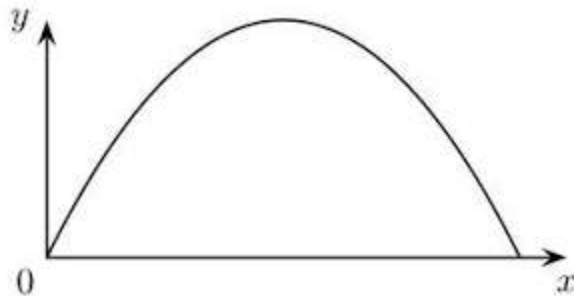
Special cases of equation of a straight line

- a) When $c = 0$, the straight line equation becomes $y = mx$. The graph will pass through the origin.
- b) When $m = 0$, the equation becomes, $y = c$.
- c) When $m = 1$ and $c = 0$, the equation becomes $y = x$
- d) When $y = 0$, it gives a line on the x-axis
- e) When $x = 0$, it gives a line on the y-axis

2.1.2. Quadratic Graph Representation

The quadratic dependence has three separate variable and so can produce a variety of parabolas. If the independent variable, x has a relationship that depends on the x^2 then it is a quadratic dependence. The equation (2) and figure 2 gives a quadratic relationship

$$y = ax^2 + bx + c \quad (2)$$



2.1.3 Exponential Graph Representation

The curves that follows the exponential decay or growth occur in many areas of Physics such as the charging and discharging of capacitors, or in the decay of radioactive material.

A general exponential growth is represented by

$$y = k \exp(ax) \quad (3)$$

A general exponential decay is written

$$y = k \exp(-ax) \quad (4)$$

Where y is the dependent variable, k is a multiplying constant, a is a positive constant and x is the independent variable.

2.1.4 Steps to consider in graph

- The **title** of the graph must be written boldly on the graph sheet
- The **axes** must be well labelled with appropriate units to show the quantity plotted
- The **scale** used must be written in one corner of the graph for both axes
- The **points** must tally with the data and the line of best fit must be picked. The line must touch at least two or three out of the points and the other points must be evenly distributed above and below the line of best fit.
- The **slope/gradient** must be indicated on the graph with either broken lines or dashed lines or solid lines. **Note**, the slope should not be worked on the graph sheet but on the graph notebook side
- The **intercept** on the graph can be read if requested

The slope of the graph can be calculated using the formula below:

$$S = \frac{y_2 - y_1}{x_2 - x_1}$$

Where y and x are variables. It is not a fixed parameter.

CHAPTER THREE

ERROR ANALYSIS

3.0 Error of Measurement

Errors are deviation of measured values from true value. For a physical quantity measurement, there is always difference between the measured value and the true value. In sequence, to know the idea of errors in measurement, one has to recognize the two conditions that describe the error namely measured value as well as true value. The measurement of error is a result of inaccuracy. Error can be measured or estimated.

There are three types of error of measurement namely;

3.0.1 Systematic error

Systematic error are errors that affect the reading in a regular and predictable manner. It may be caused by improper calibrated instrument that affects all measurements, due to the fault study of the instrument reading, due to the temperature result, force, moisture, dirt, vibration otherwise because of the electrostatic field or magnetic. The reading is consistently wrong in one direction. Example of systematic error are heat losses in calorimeter experiments, zero error in voltmeter or ammeter, approximation in theories. This can be detected by using different methods to measure quantities and comparing their results

3.0.2 Erratic errors

Erratic errors are errors that affects the reading in a regular manner. It arises from personal mistakes. Examples are wrong calculation, wrong counting of number of oscillations, wrong reading and so on. This type of error can be reduced or eliminated by regular and constant practice and by being more careful.

3.0.3 Random errors

Random errors are statistical fluctuations (in either direction) in the measured data due to the precision limitations of the measurement device. it makes the reading to be larger or smaller than the true reading. Random errors are due to fluctuations in external condition such as pressure and temperature, fluctuations in current and voltage in electricity experiment, imperfection in our senses, and mechanical vibration of the experimental set-up. Random

errors can be evaluated through statistical analysis and can be reduced by averaging over a large number of observations.

Apart from these major errors, we also have zero error, total error, relative error, percentage error and parallax error.

3.1 Error of Estimation

There are two important indices of the uncertainty in a measured quantity. They are standard and maximum errors.

3.1.1. Maximum error

This is the total error which a measured value could contain. In most experimental measurements, it is very easy to estimate a maximum error. It gives the outside limit of the error.

3.1.2 Standard error

The standard error of measurement is used to determine the effect of measurement error on individual results. The quantity which measures the amount of random fluctuations in a value is the standard deviation which is the function of standard error. The standard error of the mean is given as

$$S = \frac{W}{n}$$

Where W is the range of the value (the difference between the greatest and the least value) and n is the number of measured values.

For example: Suppose a measurement is made and the length measured are 4.7, 4.6, 4.9, 4.5, 4.8cm. The true value is

$$L = \frac{4.7 + 4.6 + 4.9 + 4.5 + 4.8}{5}$$

$$L = 4.7cm$$

And the standard error is

$$S = \frac{W}{n}$$
$$S = \frac{4.9 - 4.5}{5}$$
$$S = 0.1$$

Therefore, $L = (4.7 \pm 0.1)cm$

3.2 Combination of Error

The result in experiment is not gotten from just one measurement but from several numbers of measured quantities x, y, z with V as the final value. The maximum and standard error are represented by E_x and S_x respectively. The error can be in sum and difference, product and quotient.

3.2.1 Error in sum and difference

Given $V = x \pm y \pm z \pm \dots$

Then, the standard error is $S_V^2 = S_x^2 + S_y^2 + S_z^2 + \dots$

The maximum error is $E_V = E_x + E_y + E_z + \dots$

Example: Given $P = y - z$, such that $y = 94.0 \pm 0.25cm$ and $z = 85.5 \pm 0.25cm$. Calculate the standard and maximum error of P .

Solution: $P = y - z = 94.0 - 85.5 = 8.5$

The standard error is $E_P = E_y + E_z = 0.5 + 0.5 = 1.0$

Therefore; $P = 8.5 \pm 1.0 cm$

$$P = y - z = 94.0 - 85.5 = 8.5$$

The maximum error is $S_P^2 = S_y^2 + S_z^2 = 0.5^2 + 0.5^2 = 0.25 + 0.25$

$$S_P^2 = 0.5$$

$$S_P = \sqrt{0.5} = 0.707$$

$$\text{Therefore } P = 8.5 \pm 0.7$$

3.2.2 Errors in quotient and product

$$\text{Given } V = px \text{ or } V = px^l$$

$$\text{The standard error is } \frac{S_V}{V} = \sqrt{\left(\frac{S_P}{P}\right)^2 + \left(\frac{S_x}{x}\right)^2} \quad \text{OR} \quad \frac{S_V}{V} = \sqrt{\left(\frac{S_P}{P}\right)^2 + \left(\frac{lS_x}{x}\right)^2}$$

$$\text{The maximum error is } \frac{E_V}{V} = \frac{E_P}{P} + \frac{E_x}{x} \quad \text{OR} \quad \frac{E_V}{V} = \frac{E_P}{P} + \frac{lE_x}{x}$$

Example: If $T = 2\pi \sqrt{\frac{l}{g}}$ and $L = 100.00 \pm 0.1 \text{ cm}$ and $T = 2.00 \pm 0.01 \text{ s}$. Calculate the value of g and its standard error.

Solution:

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$T^2 = 4\pi^2 \left(\frac{l}{g}\right)$$

$$g = 4\pi^2 \left(\frac{l}{T^2}\right)$$

$$g = 4 \times (3.14)^2 \left(\frac{100}{(2.00)^2}\right)$$

$$g = 4 \times 9.87 \left(\frac{100}{4.00}\right)$$

$$g = 39.43 \times 25$$

$$g = 985.96 \text{ cm/s}^2$$

$$\frac{S_g}{g} = \sqrt{\left(\frac{S_l}{l}\right)^2 + \left(\frac{lS_T}{T}\right)^2}$$

$$\frac{S_g}{985.96} = \sqrt{\left(\frac{0.1}{100}\right)^2 + \left(\frac{2 \times 0.01}{2.00}\right)^2}$$

$$S_g = 985.96 \sqrt{\left(\frac{0.1}{100}\right)^2 + \left(\frac{2 \times 0.01}{2.00}\right)^2}$$

$$S_g = 985.96 \sqrt{\left(\frac{0.1}{100}\right)^2 + \left(\frac{2 \times 0.01}{2.00}\right)^2}$$

$$S_g = 985.96 \times \sqrt{(0.000001 + 0.005)}$$

$$S_g = 985.96 \times 0.0707$$

$$S_g = 69.70$$

The standard error for g is $985.96 \pm 69.70\text{cm}$.

The relationship between standard error, S and maximum error, E in a measured quantity is $E=3S$.

PRACTICE EXERCISES

1. Define error, hence briefly explain two types of error of estimation and two types of
2. The following repeated readings were taken: 24.3, 24.8, 24.2, 24.4, 24.6, 24.7 and 24.5. Find the standard error of these readings.
3. If $U = \frac{kl}{5m^2n}$ where $k = 1.10 \pm 0.05 \text{ cm (m.e)}$, $L = 2.03 \pm 0.01 \text{ cm (m.e)}$, $m = 1.04 \pm 0.02 \text{ (m.e)}$ and $n = 1.50 \pm 0.02 \text{ cm (m.e)}$. Calculate the error in U.
4. Calculate the maximum error in P, given $P = I^2 R$ where $I = 2.00 \pm 0.05 \text{ A (m.e)}$ and $R = 5.00 \pm 0.01 \Omega \text{ (m.e)}$.
5. If a torque is applied at the wire, the angle of rotation θ at a fixed end is given
 $mgR = \frac{na^4\theta}{2l}$, where $\theta = 0.100 \pm 0.008 \text{ rad}$; $m = 3.500 \text{ kg}$; $a = 0.005 \pm 0.0001 \text{ m}$; $R = 0.068 \pm 0.002 \text{ m}$; and $g = 9.800 \text{ ms}$. Calculate n and its standard error by neglecting m and

6.(a) With the aid of diagram, explain any two types of graph

(b) If x_1, y_1 and x_2, y_2 are two points on the line of graph as given below;

$$y_1 = mx_1 + C \text{ and } y_2 = mx_2 + C.$$

Derive the expression for slope stating the meaning of all parameters given.

(c) Calculate the maximum error in P, given $P = I^2 R$ where $I = 2.00 \pm 0.05 \text{ A (m.e)}$ and $R = 5.00 \pm 0.01 \Omega \text{ (m.e)}$.

CHAPTER FOUR

EXPERIMENT 1

SIMPLE PENDULUM

Objectives: To determine the relationship between the period of oscillation of a simple pendulum and its measured quantities and to determine the value of acceleration due to gravity.

Apparatus: Pendulum bob, retort stand with clamp, a split cork, thread/spring, metre rule and stop watch

Diagram:

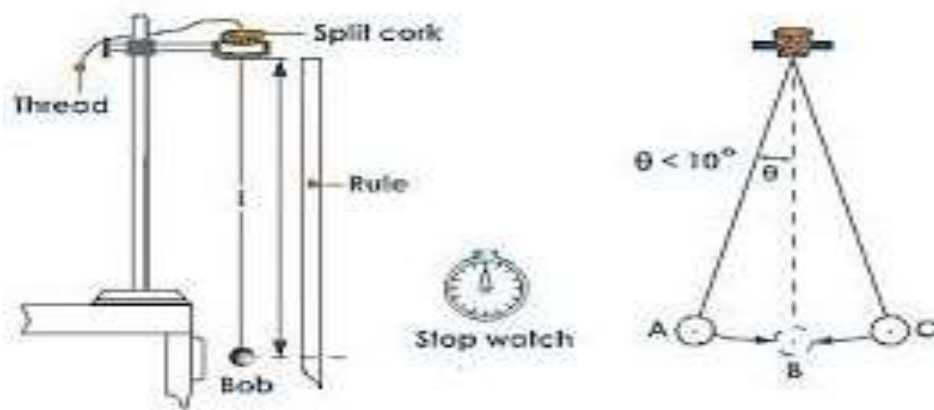


Figure 1: Simple pendulum setup

Theory: The simple pendulum consists of a point mass suspended by a string without weight. The period T of the pendulum is the time for one complete oscillation, the time for the bob to swing from C to B then back to A (See figure 1). The relation for the period of the motion is given as

$$T = 2\pi \sqrt{\frac{l}{g}}$$

Where g is the acceleration due to gravity.

Procedure:

In the diagram above, the pendulum bob is hanged on the retort stand as shown in figure 1. The angle at which the pendulum is drawn out should be small. The smaller the angle, the closer the harmonic motion the pendulum moves through instead of being outside of it. The length 20cm of the thread to the centre of gravity of the bob should be measured. With the use of stop watch, the time, t takes to complete 20 oscillations is measured. The procedure is repeated for length 40, 60, 80 and 100cm and the corresponding time.

- (a) Measure and record the time t
- (b) Plot the graph of T^2 against L
- (c) Determine the slope and intercept
- (d) From the slope and the expression $T=2\pi\sqrt{\frac{l}{g}}$, calculate the value of acceleration due to gravity g
- (e) How does your value compare to the known value of 9.8m/s^2 . Are they the same, if not why?

EXPERIMENT 2

THE VERNIER CALIPER

Objectives: To study the vernier scale principle, and to learn the use of vernier calipers for the accurate measurement of length.

Apparatus: Vernier Caliper,

Diagram:

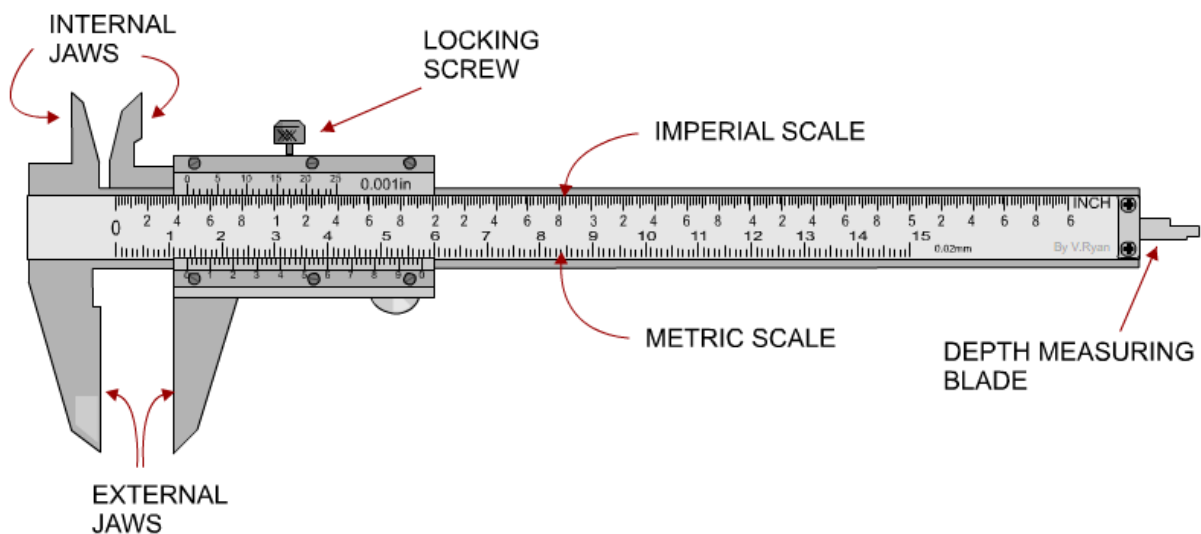


Figure 2.0

Theory: The Vernier caliper principle gives a more accurate determination of length than the metre rule. They are used for measurement about 10mm to 120mm. the caliper has two jaws, the internal jaw measures the internal dimension while the external jaw measures the external dimension. The main scale is graduated in centimetres and in millimetres. The smallest division on the main scale that can be read is 0.1mm or 0.01cm.

Procedure: Close the jaws of the calipers and make sure the main scale and Vernier scale coincide at zero. If not, note the zero error. It should be added to (if the reading is above the zero of the main scale) or subtracted from (if the reading is below the zero of the main scale) all subsequent readings. Open the jaw and placed the object to be measured in the jaw. Lock the instrument using the nut. Read the main scale at zero value of vernier scale and read the vernier scale too by looking at the point where the main and vernier scale are directly in line

with each other. The sum of the main reading and vernier reading gives the measurement of the dimension of the object placed on the jaw. Tabulate your readings.

- (a)** Write a report on vernier caliper following the format of writing report
- (b)** Use the obtained values to obtain the surface area and volume of each objects given.
- (c)** Why have you used vernier caliper to measure the objects
- (d)** How accurate can you measure the dimension of the objects with a metre rule

EXPERIMENT 3

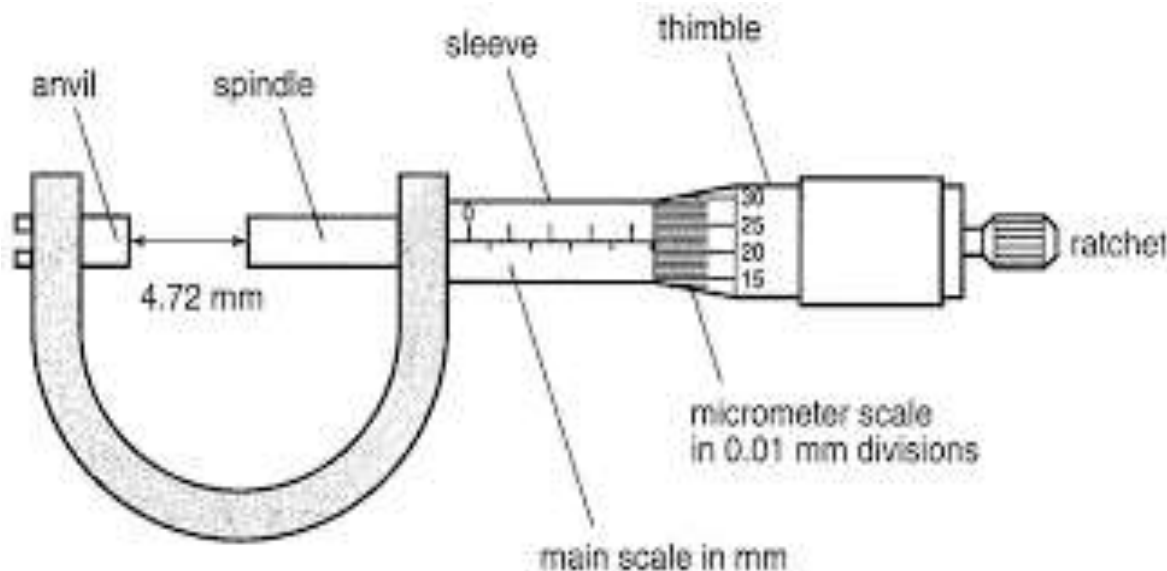
THE MICROMETER SCREW GAUGE

Objectives: To study the micrometer screw gauge principle and to learn the use of micrometer screw gauge for the accurate measurement of length.

Theory: Micrometer screw gauge is another instrument for measuring small distances and diameter which is shown in figure 3. The micrometer screw gauge has the main scale and the thimble scale with highest reading accuracy of 0.01mm. The main scale has a screw with a pitch of 0.5mm and the thimble graduated to 0.01mm revolution.

Apparatus: Micrometer screw gauge, bob, metre rule and microscope slide

Diagram:



Procedure: Carefully close the jaws of the screw using the ratchet and record the reading of the thimble scale. Check if there is zero error and take note of it. The zero error must be added or subtracted from subsequent readings. Unscrew the instrument and place the object to be measured in between the jaws and close firmly using the ratchet. Record the reading on the main scale and add that of thimble reading with it. Add or subtract the zero error if there

is any. The corresponding dimensions of different objects given are measured and recorded appropriately.

- (a)** Write a report on micrometer screw gauge following the format of writing report
- (b)** Use the obtained values to determine the surface area and volume of each objects given.
- (c)** Calculate the density of the microscope slide if the mass is 5.0g
- (d)** Why have you used micrometer screw gauge to measure the objects?
- (e)** How accurate can you measure the dimension of the objects with a metre rule

EXPERIMENT 4

HOOKE'S LAW

Objectives: The objectives are to show that the period of oscillation of a simple pendulum depends on its length and determine the value of acceleration due to gravity by means of simple pendulum.

Apparatus: spiral spring with pointer, retort stand with clamp, metre rule, slotted weights

Theory: An elastic body such as a spiral spring changes its length when force is applied to it. Hooke's law states that the change in length/extension is proportional to the force applied, then it can be expressed as

$$F = ke$$

Where k is the force or spring constant, F is the force applied and e is the extension in length.

Diagram:

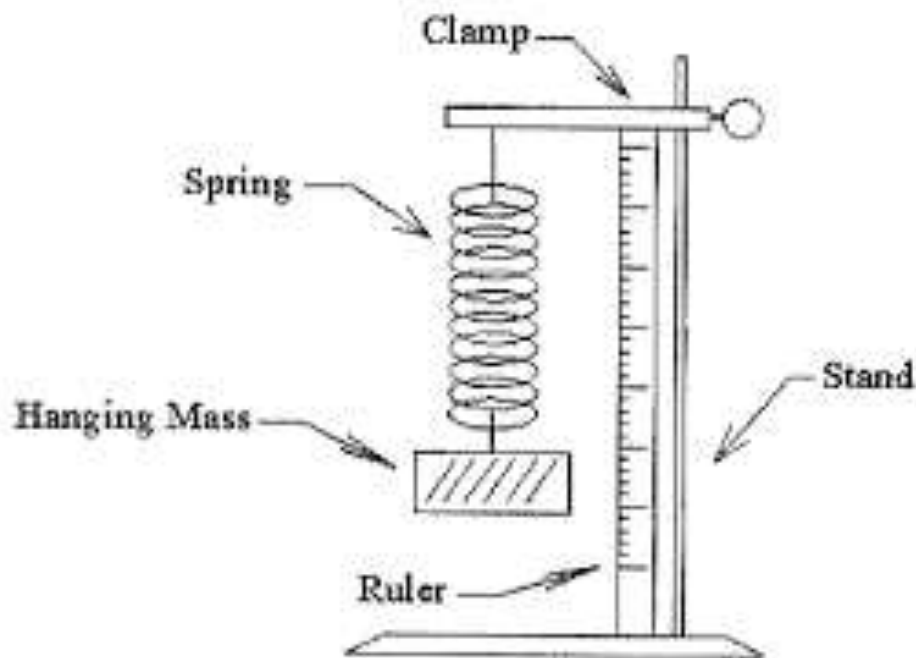


Figure 4

Procedures: Mount the metre rule and the spiral spring on the retort stand with clamp as shown in figure 4. Add a considerable mass to the spiral spring that will separate the coils of the spring and record the position of the pointer on the metre rule which is the initial length l_0 of the initial mass M_0 added. Add a mass $m = 20\text{g}$ and record the length using the pointer. Consequently add masses 40, 60, 80 and 100g and note and record their corresponding lengths L_1 . Then remove the masses one after the other and note the new lengths L_2 . Tabulate your readings as follows:

Questions

- (a) Calculate the extension, e and extending force, F
- (b) Plot the graph of F against e
- (c) Determine the slope and states its units
- (d) Does Hooke's law holds?
- (e) From your graph, calculate the value of the spring constant.
- (f) If a different spiral spring is given, what would be the relationship?

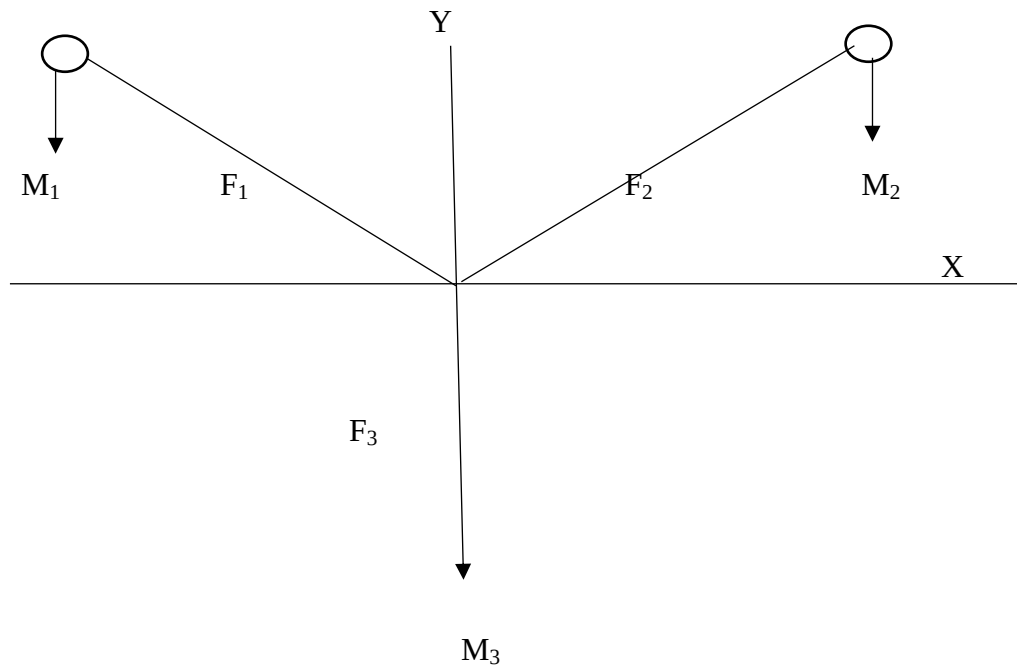
EXPERIMENT 5

THE COMPOSITION AND RESOLUTION OF FORCES

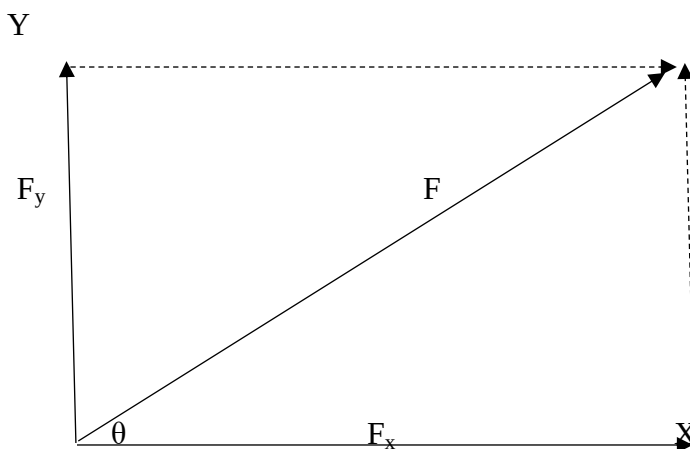
Objectives: To test the resolution of forces

Apparatus: force board with pulley, set of masses, string (thread) and A4 paper

Diagram:



Theory: The resolution method is mostly used to find the resultant of many vectors. A single vector may be resolved into two separate vectors in mutually perpendicular directions and the forces are the components of the given forces. As shown below



Using Pythagoras theorem,

$$F_x = F \cos \theta \dots\dots\dots 1$$

$$F_y = F \sin \theta \dots\dots\dots 2$$

Where θ is the angle between two forces

If there are several forces acting at a point, the X-components of the resultant R_x is equal to the sum of the X-components of the individual forces and the Y- components, R_y is equal to the sum of Y-components. That is;

$$R_x = \sum F_x \dots\dots\dots 3$$

$$R_y = \sum F_y \dots\dots\dots 4$$

The final resultant force is given as

$$R = \sqrt{R_x^2 + R_y^2} \dots\dots\dots 5$$

The directional angle with X-axis is also given as

$$\tan \theta = \frac{R_y}{R_x} \dots\dots\dots 6$$

If the forces are in equilibrium then the resultant must be equal to zero. Therefore the sum of the X-components and the sum of the y-components must be equal to zero also. That is;

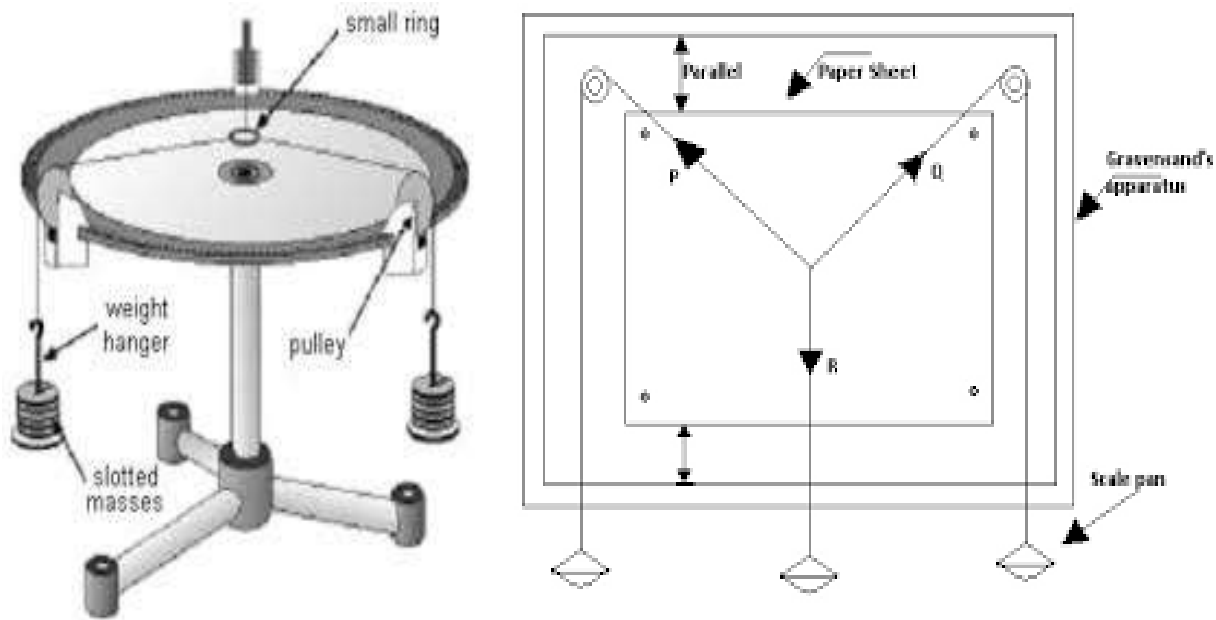
$$\sum F_x = 0 \dots\dots\dots 7$$

$$\sum F_y = 0 \dots\dots\dots 8$$

The masses suspended from strings are used to produce the forces at a point. The vectors act along the directions of the strings and are proportional to the masses hanging from the strings. These forces are represented by lines having both length and direction.

From the diagram above (Figure 5.0), F_1 acts at an angle α to the horizontal, F_2 acts at an angle β to the horizontal and F_3 acts vertically downwards.

Procedure: The known masses M_1 , M_2 , M_3 are connected by strings running over the pulley as indicated. When the equilibrium has been reached, the drawing paper should be stuck on the force board behind the point of intersection of the string. The two points should be marked and joined along the directions of the string to produce an intersection of the three lines as shown in figure 5.0. These lines are vectors representing the forces involved. With the use of ruler, the axes were constructed at a point of intersection of the three forces.



Questions

- Measure the angles α and β
- Determine the F_x and F_y
- What do the three vectors you have drawn represent?
- If a force of 50N acts at an angle of 30° to the horizontal. Calculate its horizontal and vertical components

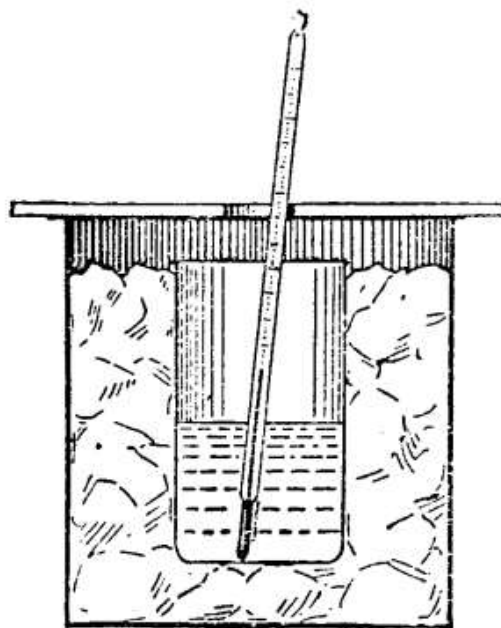
EXPERIMENT 6

THE SPECIFIC HEAT CAPACITY OF A SOLID

Objective: To determine the specific heat capacity of a metal ball using method of mixture

Apparatus: Calorimeter with stirrer, copper ball, stirrer, thermometer, heater, container and thread

Diagram:



Theory

Heat capacity of a body is the quantity of heat required to raise the temperature of the body by 1°C . The specific heat of a substance is the heat capacity per unit mass. Thus, heat capacity = mass $\times$ specific heat. The specific heat is essentially a measure of how thermally insensitive a substance is to the addition of energy. The water equivalent of a body is the mass of water, which would require the same amount of heat as the body in order to raise the temperature through one degree Celsius. The method of mixtures makes use of the principles that when two bodies at different temperatures exchange heat, the quantity of heat lost by the warmer body is equal to the heat gained by the cooler body, and some intermediate equilibrium temperature is finally reached. This is true provided no heat is lost or gained from/to the surroundings. The purpose of the calorimeter is to prevent heat loss to the surroundings. There are three methods of heat transfer: conduction, convection and radiation. In this experiment, a heated solid of known mass and temperature is dropped in the calorimeter containing known mass of cold water. The equilibrium temperature is then measured. The magnitude of the heat lost by the solid must be equal to the magnitude of the heat gained by the water, and calorimeter with stirrer. The empty

and dry clean calorimeter with stirrer is weighed. The calorimeter is half-filled with water and weighed. Boil some water in a beaker to boiling point and find the mass of the copper ball (solid). The temperature of the water should be recorded.

Procedure

Fill the beaker about half way with water and start heating it. Record the name of the metal being used in the data table. Weigh the solid (m) metal, and then lower it into the beaker of hot water by means of a thread, to avoid burns. While the solid is being heated. Weigh the inner chamber of calorimeter and the stirrer, together. Fill the inner chamber of the calorimeter about half way with cool water and add one or two small pieces of crushed ice to the water. Weigh the inner chamber of the calorimeter, stirrer and cold water ($m_1 + m_2$). Place the inner chamber of the calorimeter into the outer calorimeter jacket and place the lid on, then record the temperature of the cold water (T_2). Be sure to stir the water first. 9. Record the temperature of the hot solid when the temperature becomes steady (T_1). This should occur after the water boils. Hint: The metal should be the same temperature as the hot water. Now quickly transfer the solid from the hot water to the calorimeter without splashing any water. You must do this very quickly. 11. Place the lid onto the calorimeter and stir the water very gently and record the final equilibrium temperature (T_3). Be careful not to break the thermometer, while stirring the mixture.

Mass of the solid (metal) in kg = m

Specific heat of the metal = c

Initial temperature of (hot) solid = T_1

Mass of calorimeter with stirrer = m_1

Mass of cold water = m_2

Initial temperature of cold water = T_2

Final equilibrium temperature of mixture = T_3

Specific heat of calorimeter (Aluminum) = $C_1 = 900 \text{ J/kgC}$

Specific heat of water = $C_2 = 4186 \text{ J/kgC}$

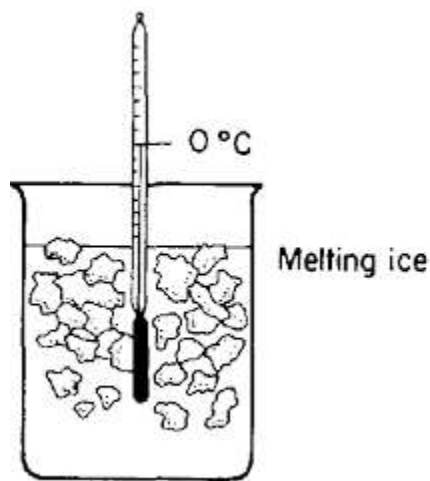
EXPERIMENT 7

THE LATENT HEAT OF FUSION IN ICE

Objective: To determine the latent heat of fusion in ice

Apparatus: unlagged calorimeter, ice pellets, beaker with water, thermometer, filter paper and weighing balance.

Diagram:



Theory: When heat is added to a substance, a temperature change is generally observed to occur. The heat added, Q , that causes a temperature change ΔT is

$$Q = mc\Delta T \quad (1)$$

where m is the mass of the substance and c is its specific heat, which is assumed to be constant over the temperature change, ΔT . However, situations exist where heat added to a substance does not cause a change in the temperature. In these cases, the added heat causes a change in phase to occur. Two common changes in phase are from solid to liquid and from liquid to gas. The amount of heat required to accomplish a phase change is called the heat of transformation. More specifically, for a solid to liquid phase change, it is referred to as the heat of fusion; and for a liquid to gas phase change, the heat of vaporization. The heat of fusion or vaporization can be expressed mathematically as

$$Q = mL \quad (2)$$

where L is the latent heat of fusion or vaporization, depending on the phase transition that occurs. In this experiment, an ice cube of mass m , assumed to be at 0°C , is placed in a calorimeter containing a mass of water m_w at temperature T_1 . After the ice cube melts, the temperature of the system is T_2 . In order to calculate the heat of fusion of ice, it is necessary to first determine the water equivalent of the calorimeter. The value m_{ccc} is found by mixing known quantities of warm water and cool water in the calorimeter. Suppose the Calorimeter contains a mass of warm water m_{ww} , at temperature T_w . If a mass of cool water m_{cw} , at temperature T_c is mixed with the warm water in the calorimeter. Thermal equilibrium will be established at an intermediate temperature T . When the heat lost and the heat gained are equated, and the resulting equation solved for the water equivalent of the calorimeter.

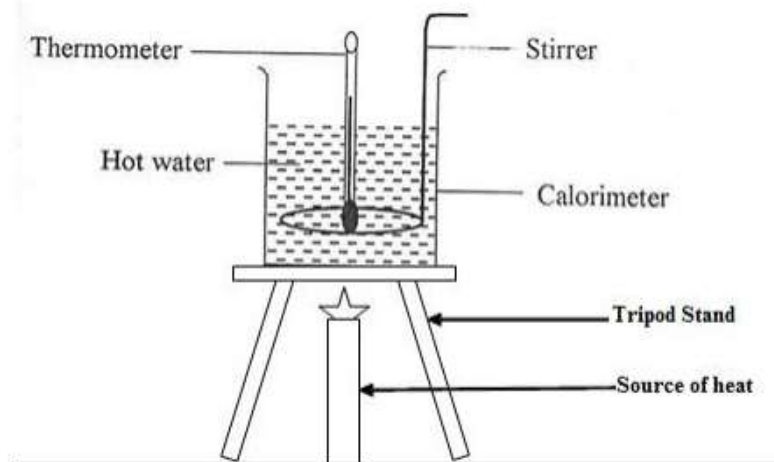
Method:

Find the mass of the empty calorimeter with the $0\text{-}50^{\circ}\text{C}$ thermometer inserted in the cork. Obtain water from the tap that is approximately 10°C above room temperature. Pour this water into the calorimeter until it is $1/3$ full. Wait for thermal equilibrium to be established, then record the temperature, T_w . Find the mass of the calorimeter with the warm water. Cool some water in a container with crushed ice until its temperature is approximately 10°C below room temperature. Record the temperature of the cool water, T_c , with the $0\text{-}100^{\circ}\text{C}$ thermometer. Pour the cool water into the calorimeter until the calorimeter is $2/3$ full. Wait for thermal equilibrium and record the equilibrium temperature T . Find the mass of the calorimeter with the added water. Calculate the water equivalent of the calorimeter. Empty the calorimeter and fill it until it is $1/3$ full of warm water that is approximately 15°C above room temperature. Wait for thermal equilibrium and record the temperature T_1 . Find the mass of the calorimeter with the warm water. Dry an ice cube and add it to the water. Be sure to agitate the water to ensure the complete melting of the ice cube. Continue to add "dried" ice cubes until the final temperature is at or near room temperature. Record the temperature T_2 , and find the mass of the calorimeter and contents. Calculate the latent heat of fusion of ice. Repeat the procedures two more times for a total of three trials.

EXPERIMENT 8

DETERMINATION OF THE ROOM TEMPERATURE OF WATER

Objective: To determine the room temperature, θ_R .



- (a) Measure the room temperature θ_0 .
- (b) Heat 100 cm^3 of water in a beaker to a temperature of about 90°C .
- (c) Transfer the hot water quickly into a calorimeter.
- (d) Place the thermometer in the hot water and start the stop clock when the temperature of the water is 65°C .
- (e) Record the temperature, θ , of the water after every two minutes for 14 minutes.
- (f) Record your results in a suitable including values of $(\theta - \theta_0)$, and $\log_{10} \theta - \theta_0$.
- (g) Plot a graph of $\log_{10} \theta - \theta_0$ against time, t .
- (h) Find the value, I , of $\log_{10} \theta - \theta_0$ when $t = 0$.
- (i) Find the antilog (I).
- (j) Calculate the temperature of the surroundings, θ_R , using the expression;

$$\text{Antilog (I)} = 65 - \theta_R.$$

EXPERIMENT 9

RELATIONSHIP BETWEEN MASS AND PRESSURE

Objective: To explore how varying masses affects the pressure within a closed system.

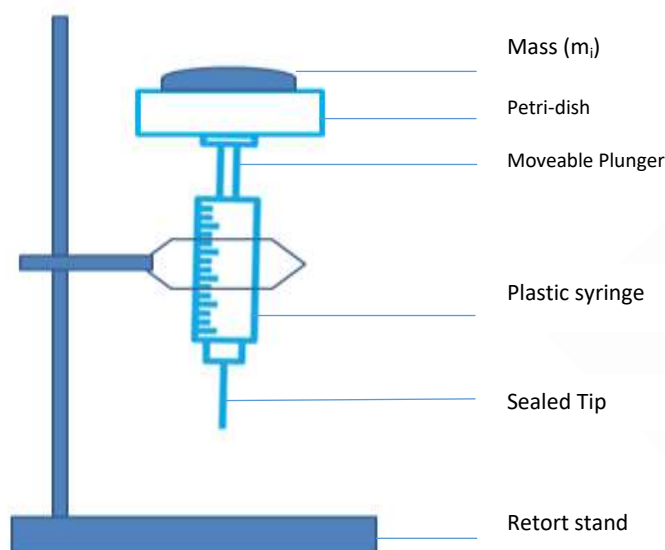


Fig (a)



Fig (b)

The diagram above illustrates a syringe with a petri-dish fixed to the top of its plunger. A known mass, M , is placed on the petri-dish. The piston of the syringe moves downward, and the volume, V_i , of air beneath the piston is read and recorded. The procedure is repeated for four other values of M .

Fig (a) represents the masses, M_i , placed on the petri-dish.

Fig (b) represents the corresponding volumes, V where $i = 1, 2, 3, 4$ and 5 .

1. (i) Read and record the values of M
- (ii) Read and record the corresponding values of V
- (iii) Evaluate V^{-1}
- (iv) Tabulate your readings.
- (v) Plot a graph with M on the vertical axis and V^{-1} on the horizontal axis
- (vi) Determine the slope, s , of the graph.
- (vii) What physical law does this experiment verify?
- (viii) State two precautions that are necessary to ensure accurate results when performing this experiment

2. (i) Explain the effect of decreasing the volume of a fixed mass of gas at constant temperature using the kinetic molecular theory of gas.
- (ii) A cylinder is filled with a gas at 27°C . If the volume of the gas is increased at constant pressure to 1.25 times its original value, calculate the final temperature of the gas.

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